



Saumitra Mazumder



SAMazumder



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About Me

Experienced in all three lines of defense. Solid knowledge of a range of quantitative finance, risk management (e.g., VaR/Expected Shortfall for market risk, credit portfolio models, stress testing), derivatives models & pricing theory, as well as relevant regulatory frameworks. Interested in the intersection of machine learning, quantitative finance, and climate change.

Thorough and analytical with demonstrated leadership experience. Excellent verbal and written communication skills. Strong ability to collaborate effectively with diverse individuals and strengthen relationships to design and achieve effective solutions.

Education

M.Sc. Applied Mathematics
Toronto Metropolitan University
2020-2022

B.Sc. Mathematics and Economics
Toronto Metropolitan University
2015-2019

Knowledge

AI	Artificial Intelligence, Machine Learning, Generative AI, Large Language Models (LLMs), Agentic AI, AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Workflows, Model Evaluation, Embeddings, Fine-Tuning, and AI-assisted Data Analysis.
Model Building	Climate-adjusted credit/market risk models, borrower risk rating models, generalized additive regression models, ASFR structural models, gradient boosting, Monte Carlo methods, & time series analysis, Risk Measures (VaR, cVaR, ES, coherent and convex), Elliptical Copulas.
Tools	Python (XGBoost, OpenAI API), R (Dplyr, QRM, xts, dynlm), SAS, MATLAB, LaTeX, SQL, Java, & C++.

Recent Employment History

2026-Current	PwC, LLP - Manager Financial Services Risk and Regulatory Consultant Development Lead developer of GenAI tools for FSR&R. Python, C, C++, Java, typescript, skills.md, tools/MCP servers.
2023-2026	PwC, LLP - Senior Associate Corporate Financial Risk Management Advisory Development Led the development of implementing GenAI into a semi-automated model validation process. Experienced in building quantitative models across credit and market risk, as well as design and implementation of climate risk models (credit risk and market risk) under IPCC RCP/SSP scenarios, with heat map generation (physical risk and transition risk) and integration of climate-adjusted credit and market risk parameters into default risk frameworks. Development lead experience includes: (i) BRR/FRR models, (ii) Machine learning based scam & fraud models, (iii) Rules based AML models, (iv) Credit risk models, including low default portfolios, as well as both retail & wholesale portfolios, (v) Scenario-based analysis supporting valuation of complex exposures and stress testing under adverse market and credit conditions. Validation Extensive experience leading validation of quantitative models in compliance with regulatory and internal governance standards. Validation work spans credit, market, and liquidity risk models, including independent assessment of conceptual soundness, data quality, calibration, and performance monitoring. Validation lead experience includes: (i) IFRS 9/CECL ECL model validation for wholesale portfolios (Commercial, Corporate/Sovereign, and CRE), (ii) Valuation model validation, ensuring fair value estimation methodologies align with observable market inputs, (iii) Liquidity and stress testing frameworks, including cash flow forecasting, funding concentration risk, and market downturn stress scenarios, (iv) Fraud and AML model validation, assessing machine learning conceptual soundness under limited model development documentation and ensuring robustness and interpretability.